



IBM Developer  
SKILLS NETWORK

# Winning Space Race with Data Science

Abdujabborov  
Abdurasul  
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# Outline

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- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix

# Executive Summary

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## Strategic Insights

- **Cost Efficiency:** Reuse of boosters drastically reduced launch costs, making SpaceX the market leader.
- **Market Expansion:** Transition from NASA cargo contracts → commercial satellites → mega-constellations → crewed missions.
- **Reliability:** Mission success rate improved from ~70% (2010–2014) to **>95% post-2017**.
- **Innovation:** Continuous iteration (grid fins, landing legs, Block upgrades) enabled operational breakthroughs.

## Executive Takeaways

- SpaceX transformed from a **new entrant (2010)** into the **dominant global launch provider (2020)**.
- **Reusability is the cornerstone** of its strategy, with Block 5 proving sustainable multi-flight operations.
- The **Starlink program** shifted payload dynamics, pushing Falcon 9 to its maximum capacity and ensuring consistent launch cadence.
- SpaceX's **operational maturity** is evident in its ability to balance commercial, government, and internal missions while maintaining high reliability.

# Introduction

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This project compiles and analyzes **SpaceX Falcon 9 launch data** from multiple sources. The goal is to understand the evolution of SpaceX's launch operations, focusing on **payload capacity, orbital destinations, booster reuse, landing success rates, and mission outcomes**.

The data spans **2010–2020**, covering the transition from early experimental flights to high-frequency, reusable operations supporting commercial, government, and internal missions (e.g., Starlink).

Key takeaways:

**Launch cadence:** From experimental → industrial scale.

**Payloads:** Grew 30x in mass, diversified across orbits.

**Reusability:** Achieved operational maturity, drastically reducing costs.

**Reliability:** Mission success rate >95% post-2017.

**Strategic impact:** SpaceX became the **dominant global launch provider** by 2020.



Section 1

# Methodology

# Methodology

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## Executive Summary

- Data collection methodology:
  - Describe how data was collected
- Perform data wrangling
  - Describe how data was processed
- Perform exploratory data analysis (EDA) using visualization and SQL
- Perform interactive visual analytics using Folium and Plotly Dash
- Perform predictive analysis using classification models
  - How to build, tune, evaluate classification models

# Data Collection

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- Describe how data sets were collected.
- You need to present your data collection process use key phrases and flowcharts

# Data Collection – SpaceX API

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- The data was collected through SpaceX API.
- Requested and parsed the SpaceX launch data using the GET request
- Filtered the dataframe to only include Falcon 9 launches
- Data wrangling: dealt with missed values
- [https://github.com/abdujabborovabdurasul16-source/The-research-project/blob/main/jupyter-labs-spacex-data-collection-api-v2%20\(1\).ipynb](https://github.com/abdujabborovabdurasul16-source/The-research-project/blob/main/jupyter-labs-spacex-data-collection-api-v2%20(1).ipynb) - data collection process.

```
spacex_url="https://api.spacexdata.com/v4/launches/past"
```

```
response = requests.get(spacex_url)
```



# Data Collection - Scraping

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The Wikipedia was used in webscraping process:

Requested the Falcon9 Launch Wiki page from its URL

Extracted all column/variable names from the HTML table header

Created a data frame by parsing the launch HTML tables

The source: <https://github.com/abdujabborovabdurasul16-source/The-research-project/blob/main/jupyter-labs-webscraping.ipynb>

```
static_url = "https://en.wikipedia.org/w/index.php?title=List_of_Falcon_9_and_Falcon_Heavy_launches&oldid=1027686922"

headers = {
    "User-Agent": "Mozilla/5.0 (Windows NT 10.0; Win64; x64) "
                  "AppleWebKit/537.36 (KHTML, like Gecko) "
                  "Chrome/91.0.4472.124 Safari/537.36"
}
```

# Data Wrangling

- First of all, I calculated the number of launches on each site
- Next, the number and occurrence of each orbit was determined
- After that, I calculated the number and occurrence of mission outcome of the orbits
- I created a landing outcome label from Outcome column

| ReusedCount | Serial | Longitude  | Latitude  | Class |
|-------------|--------|------------|-----------|-------|
| 0           | B0003  | -80.577366 | 28.561857 | 0     |
| 0           | B0005  | -80.577366 | 28.561857 | 0     |
| 0           | B0007  | -80.577366 | 28.561857 | 0     |

# EDA with Data Visualization

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- Visualized the relationship between Flight Number and Launch Site using catplot.
- Visualized the relationship between Payload Mass and Launch Site using scatter point chart.
- Visualized the relationship between success rate of each orbit type using bar chart.
- Visualize the relationship between FlightNumber and Orbit type; Payload Mass and Orbit type; the launch success yearly trend using line chart and scatter point charts. At the end, performed feature engineering and created dummy variables
- <https://github.com/abdujabborovabdurasul16-source/The-research-project/blob/main/edadataviz.ipynb>

# EDA with SQL

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- SpaceX SQL Tasks (Concise)
- Show **unique launch sites**.
- Display **5 records** where sites start with "CCA".
- Compute **total payload mass** for **NASA (CRS)** launches.
- Find **average payload mass** for **F9 v1.1** boosters.
- List **date of first successful ground pad landing**.
- Identify boosters with **successful drone ship landings** and payload **4000–6000**.
- Count **success vs. failure mission outcomes**.
- Use a **subquery** to find booster versions with **maximum payload mass**.
- Show **month names, failed drone ship landings, booster versions, and sites** for **2015**.
- Rank **landing outcomes** between **2010-06-04** and **2017-03-20** in **descending order**.

[https://github.com/abdujabborovabdurasul16-source/The-research-project/blob/main/jupyter-labs-eda-sql-coursera\\_sqllite.ipynb](https://github.com/abdujabborovabdurasul16-source/The-research-project/blob/main/jupyter-labs-eda-sql-coursera_sqllite.ipynb)

# Build an Interactive Map with Folium

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- **Markers:** Placed at launch site coordinates, labeled with site names, and color-coded (green for success, red for failure) to show outcomes.
- **Circles:** Drawn around sites to highlight their location and make them visually distinct.
- **Lines (PolyLine):** Added between launch sites and nearby features (coastline, city, railway, highway) to show proximity and calculate distances.
- **MousePosition plugin:** Used to capture coordinates interactively when exploring the map.
- **Why added:** These objects transform raw latitude/longitude data into an intuitive visualization, making it easy to identify site locations, compare success rates, and analyze geographical relationships such as closeness to coastlines or infrastructure.
- [https://github.com/abdujabborovabdurasul16-source/The-research-project/blob/main/lab\\_jupyter\\_launch\\_site\\_location.ipynb](https://github.com/abdujabborovabdurasul16-source/The-research-project/blob/main/lab_jupyter_launch_site_location.ipynb)



# Build a Dashboard with Plotly Dash

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- Plots/Graphs Added
- **Pie Chart (Success by Site):**
  - Shows either total successful launches across all sites or success vs. failure for a selected site.
- **Scatter Plot (Payload vs. Outcome):**
  - Displays payload mass against launch outcome, colored by booster version category.
- Interactions Added
- **Dropdown (Launch Site Selection):**
  - Lets users choose “All Sites” or a specific site to filter both charts.
- **RangeSlider (Payload Range):**
  - Allows users to select a payload mass range, dynamically filtering the scatter plot.
- These all make the dashboard both **exploratory and analytical**: users can see overall success patterns and drill down into payload-specific outcomes.
- <https://github.com/abdujabborovabdurasul16-source/Research-project-in-astronomy/blob/main/spacex-dash-app.py>

# Predictive Analysis (Classification)

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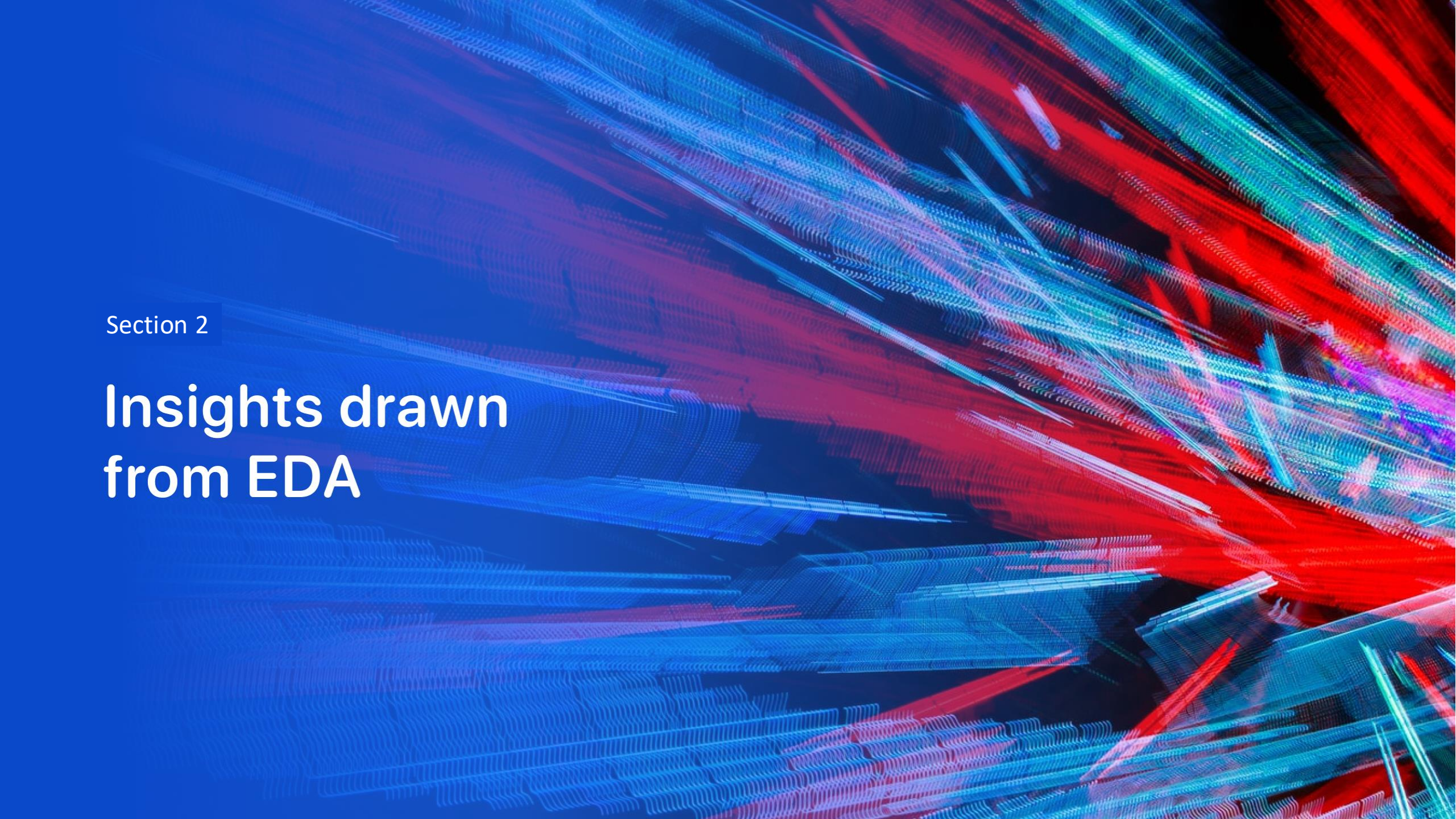
- Model Development Process
- Data Preparation
- Model Training & Hyperparameter Tuning
  - Logistic Regression
  - Support Vector Machine (SVM)
  - Decision Tree
  - K Nearest Neighbors (KNN)
  - Evaluation
    - Used `.score()` to calculate test accuracy for each model.
    - Plotted confusion matrices to analyze misclassifications (e.g., false positives in logistic regression).
- Improvement & Selection
  - Each model was tuned with **GridSearchCV** to optimize hyperparameters.
  - Decision Tree achieved the **highest validation accuracy (0.875)** but lower test accuracy (0.778), suggesting possible overfitting.
  - Logistic Regression, SVM, and KNN all achieved balanced performance (~0.833 test accuracy).
  - Conclusion
    - The process followed a structured pipeline: **data preparation → model training → hyperparameter tuning → evaluation → comparison.**
- **Best validation performer:** Decision Tree (0.875).
- **Most consistent test performers:** Logistic Regression, SVM, and KNN (~0.833).
- This shows how systematic tuning and evaluation helped identify the **best-performing classification models** while highlighting trade-offs between validation accuracy and generalization.
- [https://github.com/abdujabborovabdurasul16-source/The-research-project/blob/main/SpaceX\\_Machine%20Learning%20Prediction\\_Part\\_5.ipynb](https://github.com/abdujabborovabdurasul16-source/The-research-project/blob/main/SpaceX_Machine%20Learning%20Prediction_Part_5.ipynb)

# Results

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- Key EDA Insights
- Launch cadence accelerated dramatically after 2017.
- Payload mass distribution shifted from small ISS resupply missions to massive Starlink deployments.
- Orbit diversity reflects SpaceX's transition from NASA contracts → commercial satellites → mega-constellations.
- Booster reuse is the defining feature of Block 5, with double-digit reuses achieved.
- Landing success rates improved from <50% to >90%, enabling cost efficiency.
- SpaceX matured into a **high-reliability, high-frequency launch provider** by 2020.
- **Best validation performer:** Decision Tree (0.875).
- **Most consistent test performers:** Logistic Regression, SVM, and KNN (~0.833).
- **Trade-off observed:** Higher validation accuracy did not always translate to better test accuracy (Decision Tree overfit).



The background of the slide is an abstract composition. It features a dark blue base color. Overlaid on this are numerous diagonal streaks in shades of blue and red, creating a sense of motion or data flow. A faint, light blue grid pattern is also visible, particularly in the lower-left quadrant. The overall effect is high-tech and digital.

Section 2

# Insights drawn from EDA

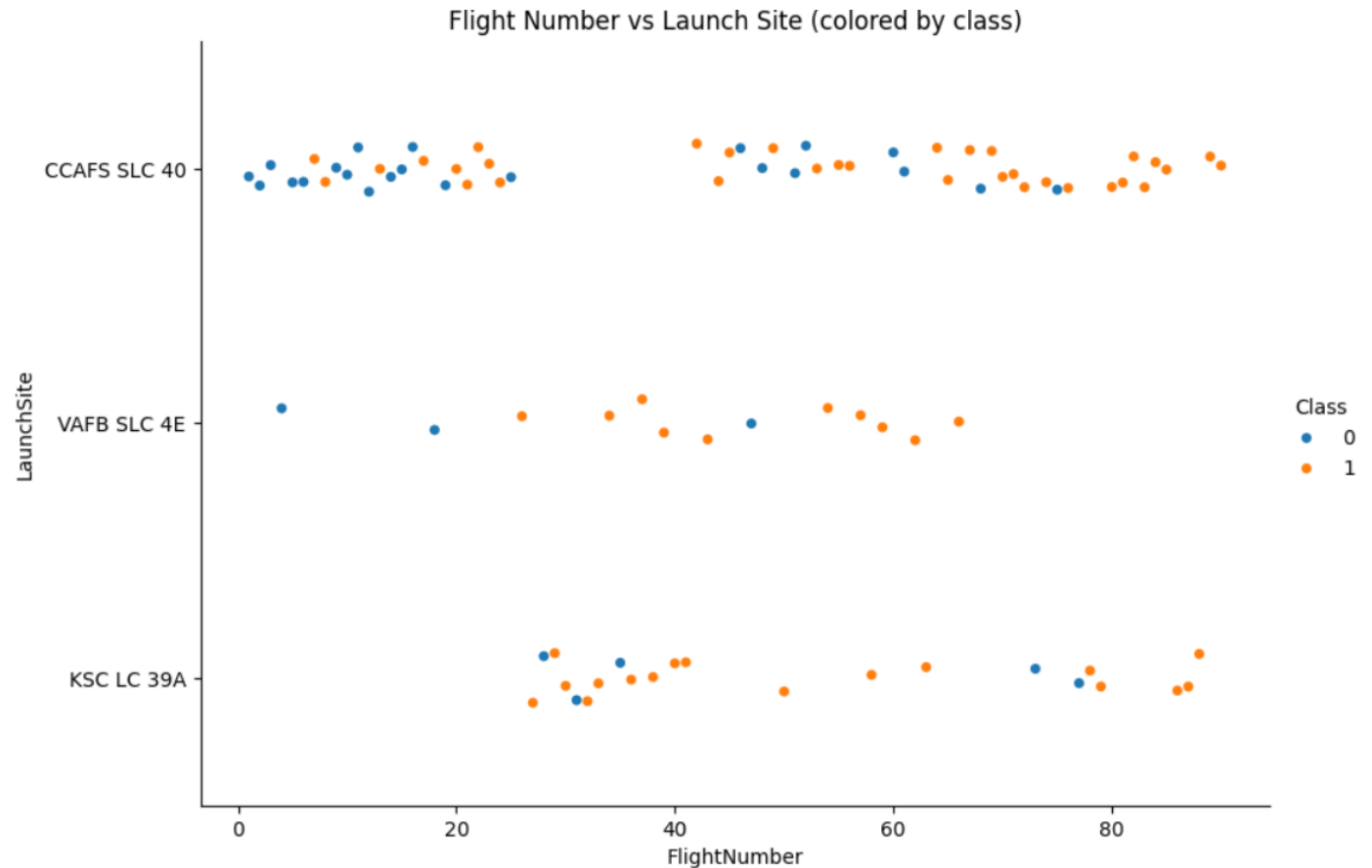


# Flight Number vs. Launch Site

The higher flight number is, the more likely it was successful.

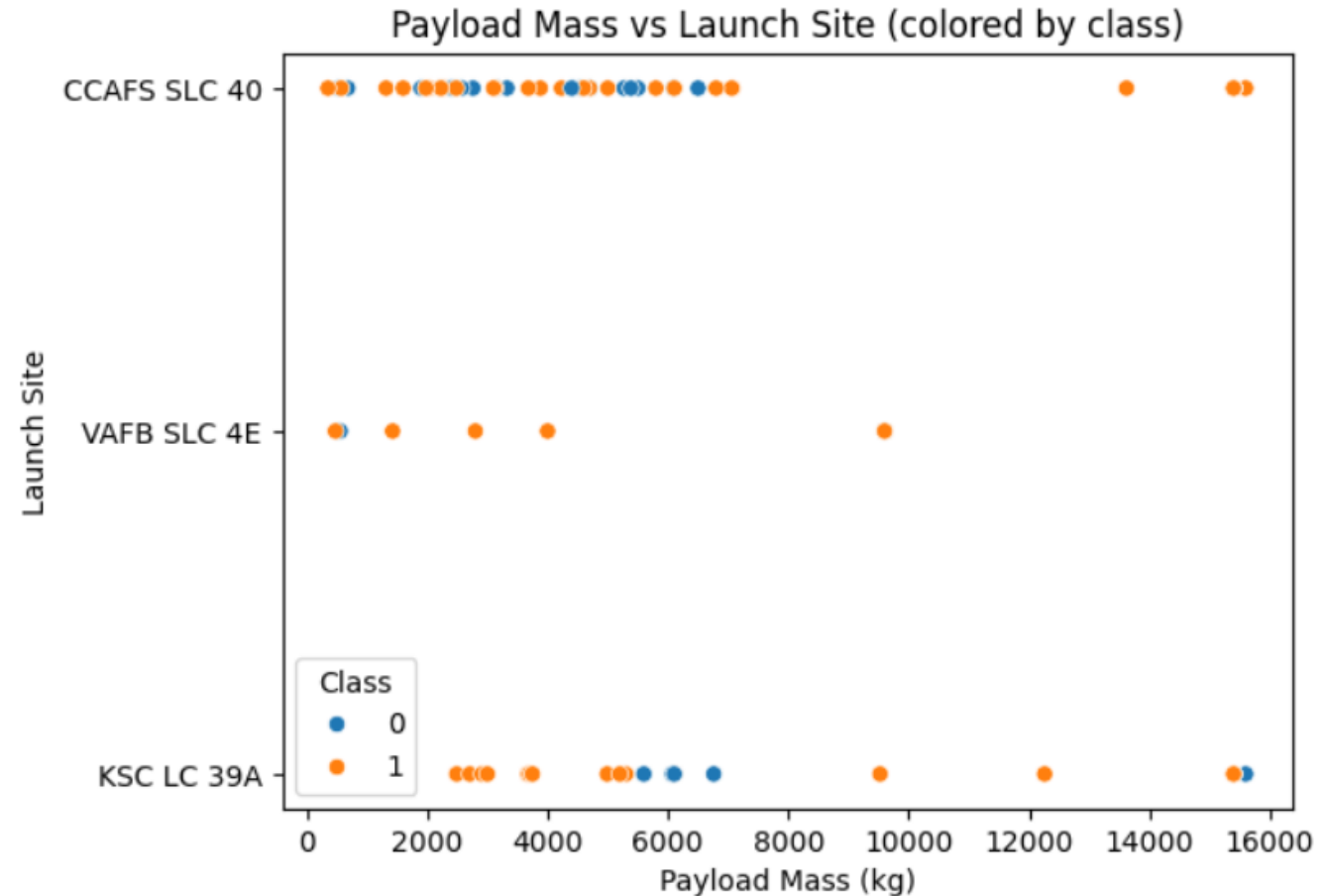
Key insights:

If the flight number is 80+, it is almost certainly successful.



# Payload vs. Launch Site

There isn't enough data to tell about VAFB SLC 4E Site and there's no clear correlation with KSC LC 39A Site. However, CCAFS SLC 40 shows that if Payload Mass is higher than 8000, it is almost certainly successful flight.

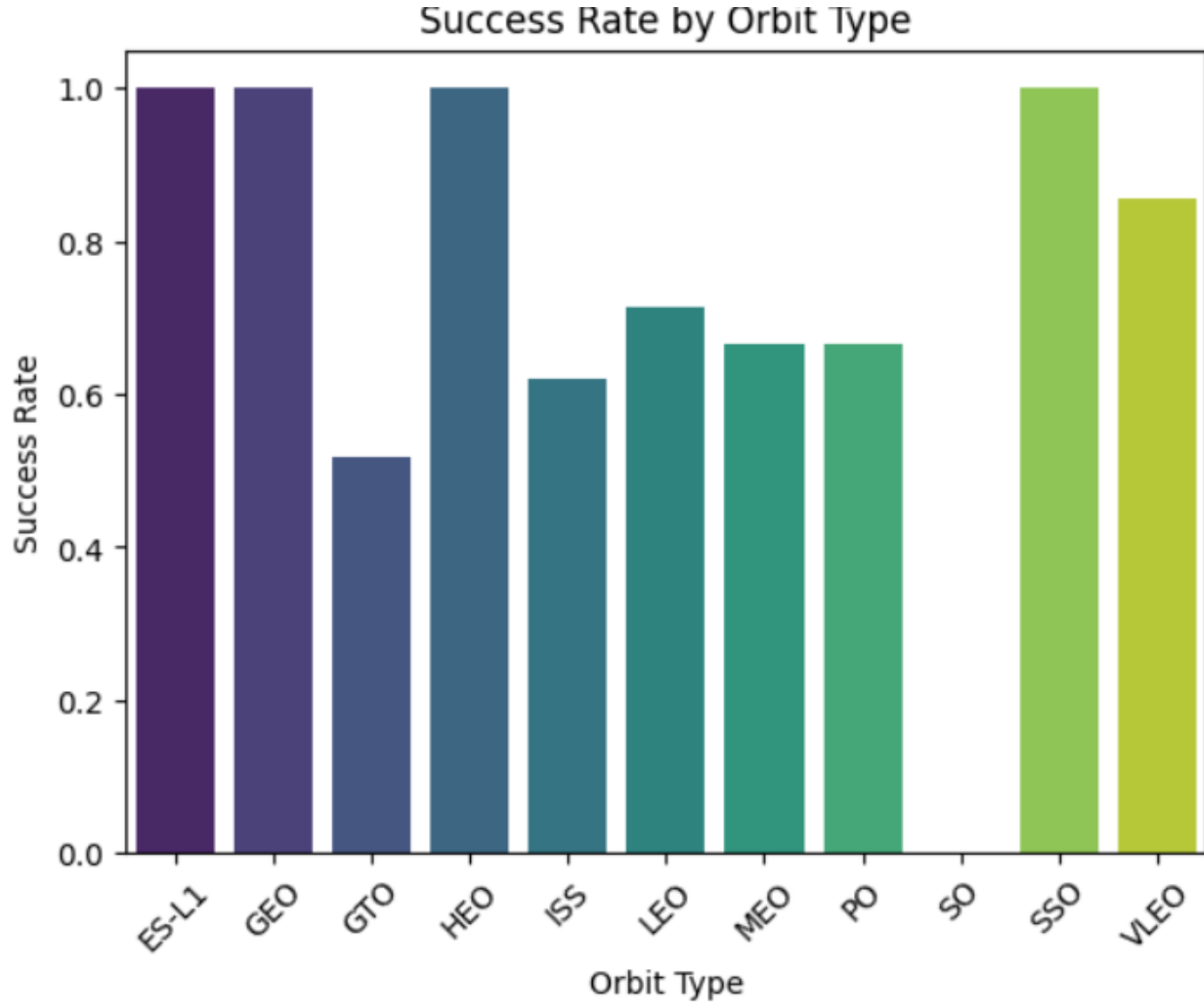




## Success Rate vs. Orbit Type

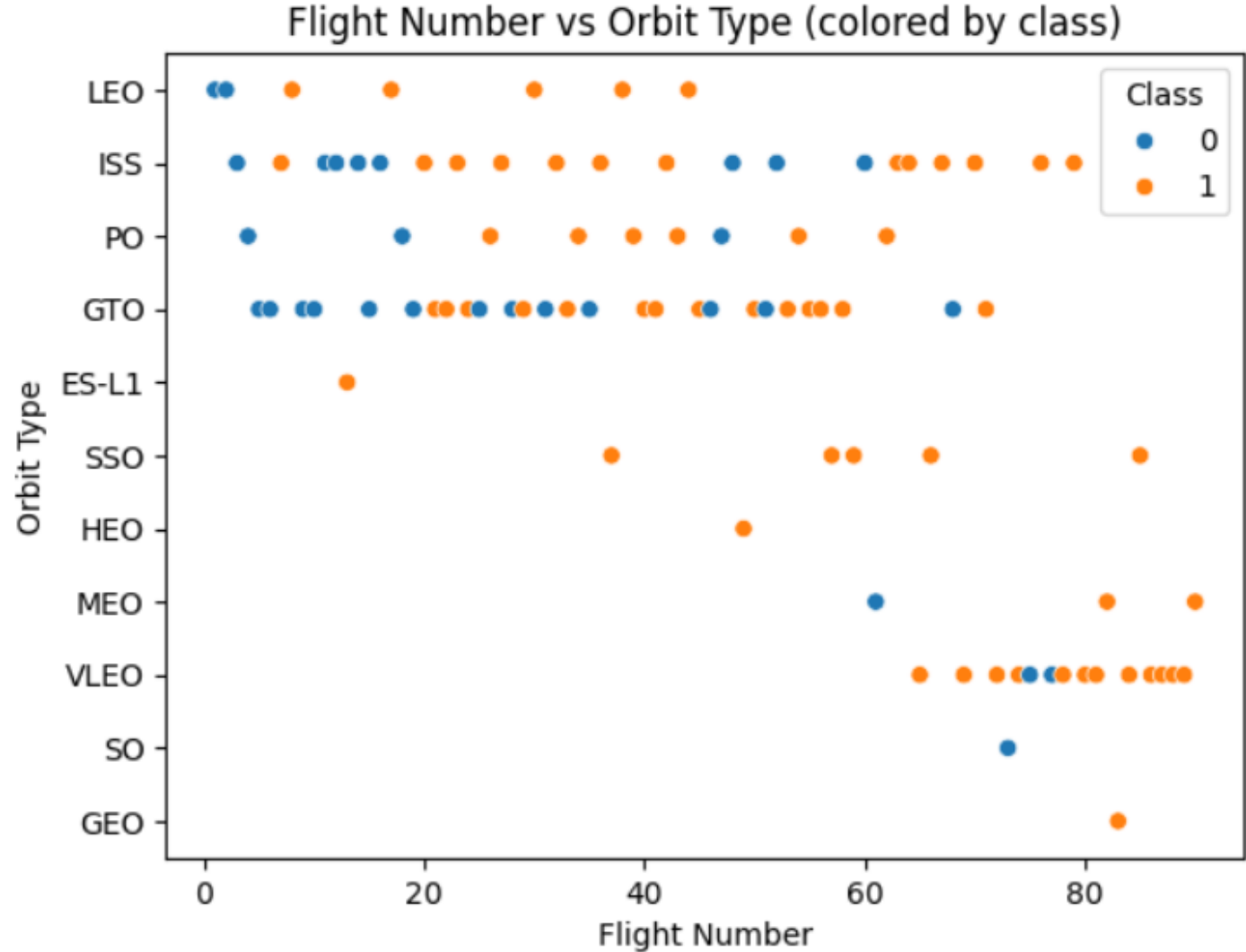
ES-L1, GEO, HEO and SSO orbits have 100% success rate. This is because there were not a lot of flights, compared to other orbits.

Meanwhile SO orbit has zero success. This can be because there was only one flight.



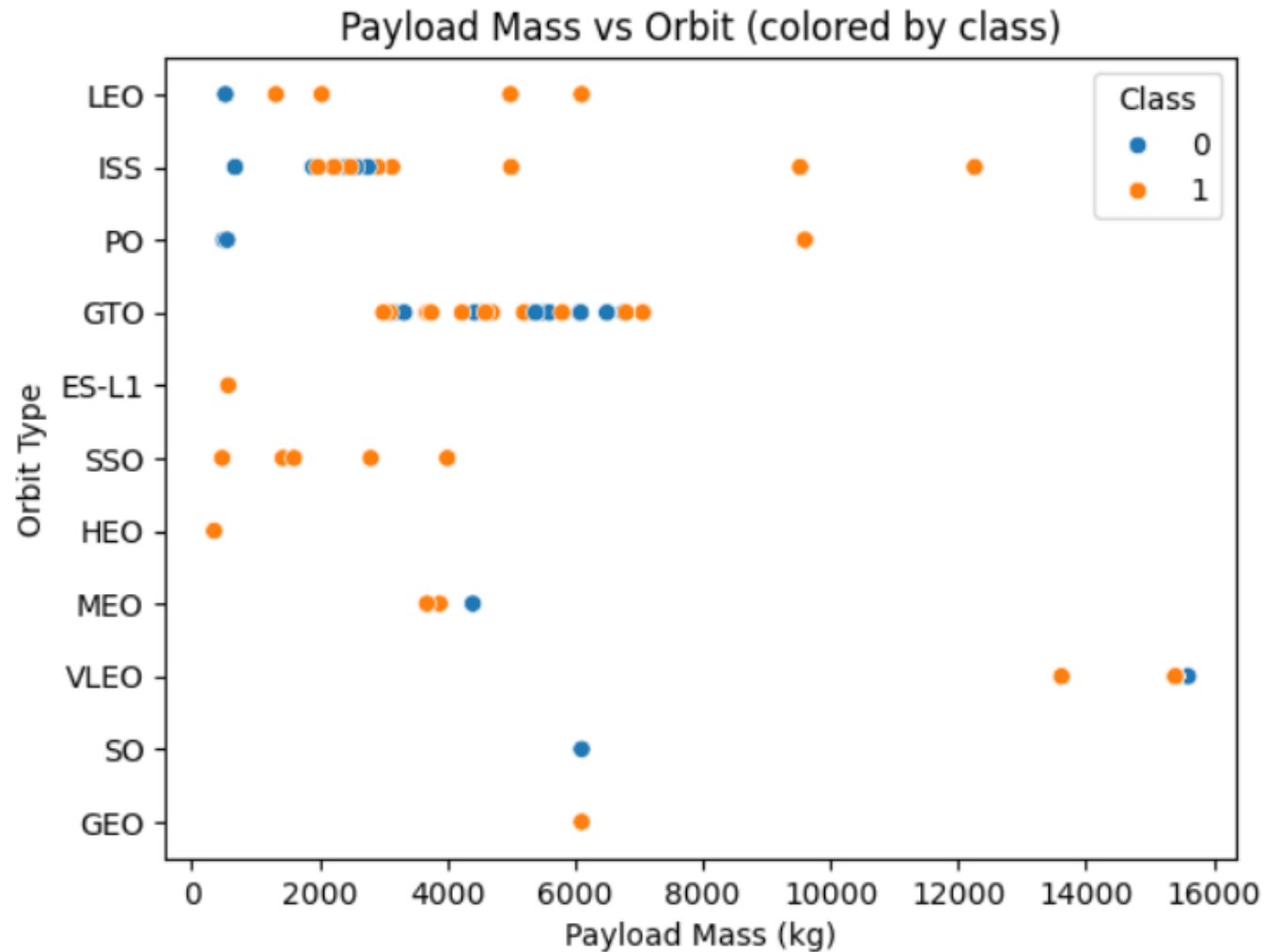
# Flight Number vs. Orbit Type

This chart provides further explanation of previous one. It shows that the more flights the orbit have, the more unsuccessful flights appear. Also, if flight number is 80+, it is certainly successful flight.



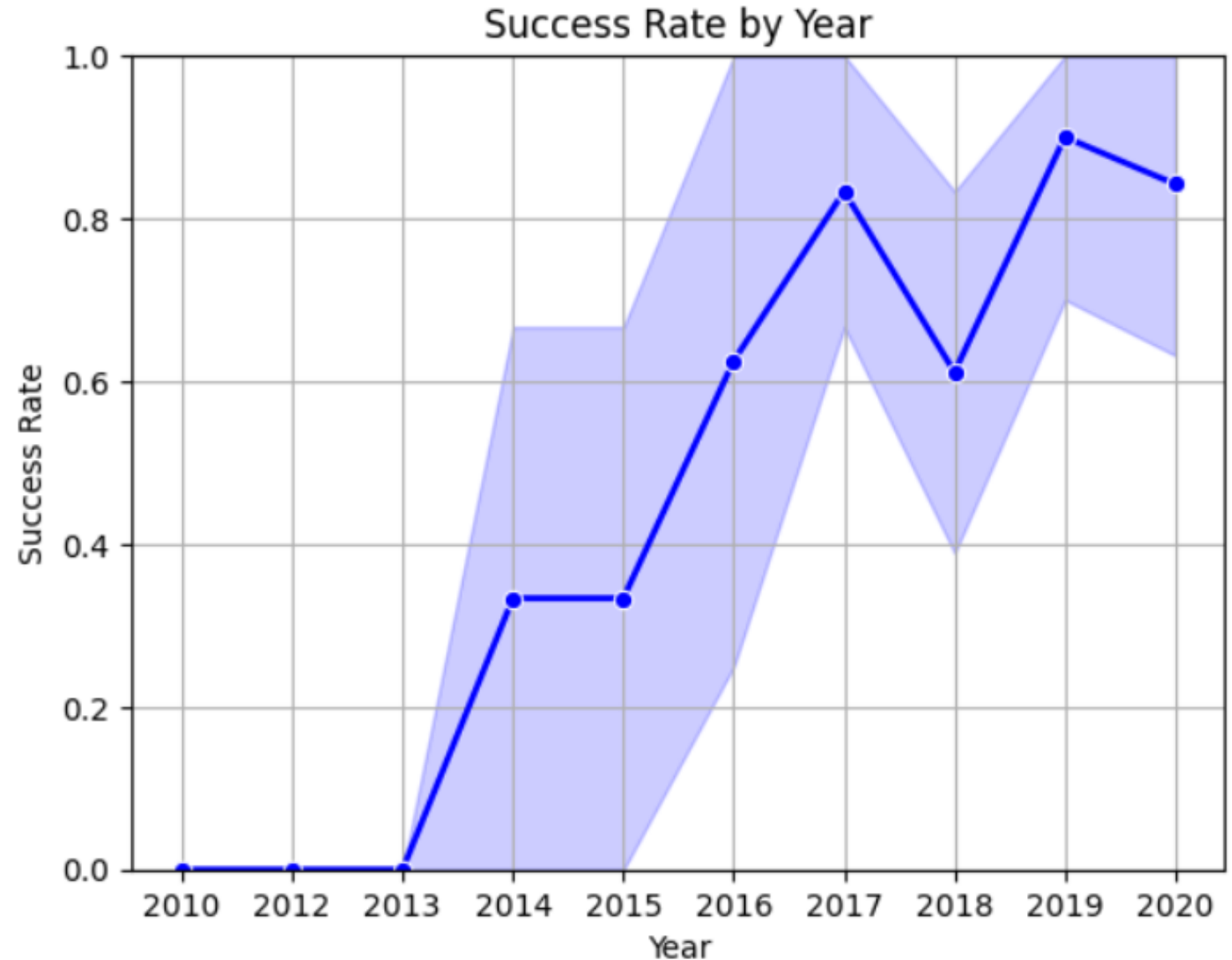
# Payload vs. Orbit Type

This scatter plot shows that all unsuccessful flights have payload mass of less than 8000, with only one exception, which has the highest payload mass. It suggests that payload mass should be in reasonable range – 8K-16K.



## Launch Success Yearly Trend

There is a clear increase in success rate throughout the 2013-2020 period, which suggests that modern technology and past mistakes have real impact.



# All Launch Site Names

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There are only 4 distinct launch sites, which are:

CCAFS LC-40

VAFB SLC-4E

KSC LC-39A

CCAFS SLC-40

SpaceX originally operated four main orbital launch sites

## Launch\_Site

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CCAFS LC-40

VAFB SLC-4E

KSC LC-39A

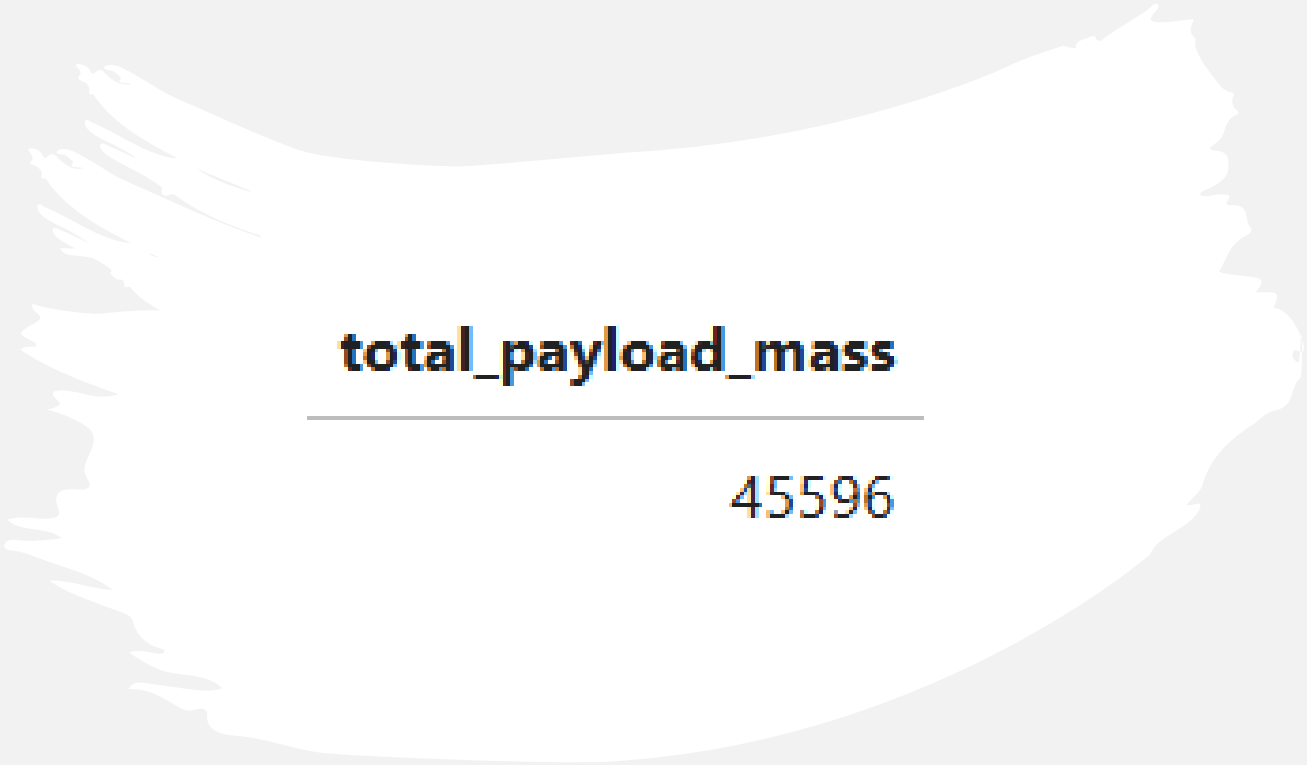
CCAFS SLC-40



## Launch Site Names Begin with 'CCA'

All five rows have launch site CCAFS LC-40.  
Even though there are some other launch sites starting with CCA, these ones appear first in the dataset

| Date       | Time (UTC) | Booster_Version | Launch_Site | Payload                                                       |
|------------|------------|-----------------|-------------|---------------------------------------------------------------|
| 2010-06-04 | 18:45:00   | F9 v1.0 B0003   | CCAFS LC-40 | Dragon Spacecraft Qualification Unit                          |
| 2010-12-08 | 15:43:00   | F9 v1.0 B0004   | CCAFS LC-40 | Dragon demo flight C1, two CubeSats, barrel of Brouere cheese |
| 2012-05-22 | 7:44:00    | F9 v1.0 B0005   | CCAFS LC-40 | Dragon demo flight C2                                         |
| 2012-10-08 | 0:35:00    | F9 v1.0 B0006   | CCAFS LC-40 | SpaceX CRS-1                                                  |
| 2013-03-01 | 15:10:00   | F9 v1.0 B0007   | CCAFS LC-40 | SpaceX CRS-2                                                  |



**total\_payload\_mass**

45596

## Total Payload Mass

The total payload mass for NASA that launched boosters

# Average Payload Mass by F9 v1.1

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The average payload mass in kilograms is around 2534.67 for F9 v1.1 version of booster

```
%sql SELECT AVG("PAYLOAD_MASS__KG_") FROM SPACEXTABLE WHERE "Booster_Version" LIKE "F9 v1.1%"
```

```
* sqlite:///my_data1.db
```

```
Done.
```

```
AVG("PAYLOAD_MASS__KG_")
```

```
2534.6666666666665
```

# First Successful Ground Landing Date

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The first ever successful landing achieved in 1st of May in 2017 by KSC LC-39A, which had the following statistics:

|            |          |               |            |         |      |     |     |         |                      |
|------------|----------|---------------|------------|---------|------|-----|-----|---------|----------------------|
| 2017-05-01 | 11:15:00 | F9 FT B1032.1 | KSC LC-39A | NROL-76 | 5300 | LEO | NRO | Success | Success (ground pad) |
|------------|----------|---------------|------------|---------|------|-----|-----|---------|----------------------|

## Successful Drone Ship Landing with Payload between 4000 and 6000

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|            |          |               |             |                       |      |     |                        |         |                      |
|------------|----------|---------------|-------------|-----------------------|------|-----|------------------------|---------|----------------------|
| 2016-05-06 | 5:21:00  | F9 FT B1022   | CCAFS LC-40 | JCSAT-14              | 4696 | GTO | SKY Perfect JSAT Group | Success | Success (drone ship) |
| 2016-08-14 | 5:26:00  | F9 FT B1026   | CCAFS LC-40 | JCSAT-16              | 4600 | GTO | SKY Perfect JSAT Group | Success | Success (drone ship) |
| 2017-10-11 | 22:53:00 | F9 FT B1031.2 | KSC LC-39A  | SES-11 / EchoStar 105 | 5200 | GTO | SES EchoStar           | Success | Success (drone ship) |
| 2017-03-30 | 22:27:00 | F9 FT B1021.2 | KSC LC-39A  | SES-10                | 5300 | GTO | SES                    | Success | Success (drone ship) |

You can see above all successful drone ship landings that have payload mass between 4K and 6K kilograms. They were achieved by CCAFS LC-40 and KSC LC-39A sites

| Total_No_Of_Success | Total_No_Of_Failures |
|---------------------|----------------------|
| 98                  | 3                    |

## Total Number of Successful and Failure Mission Outcomes

- As can be seen above, there were more successful outcomes than failed ones, which indicates high progress

30



# Boosters Carried Maximum Payload

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| Booster_Version |
|-----------------|
| F9 B5 B1048.4   |
| F9 B5 B1049.4   |
| F9 B5 B1051.3   |
| F9 B5 B1056.4   |
| F9 B5 B1048.5   |
| F9 B5 B1051.4   |
| F9 B5 B1049.5   |
| F9 B5 B1060.2   |
| F9 B5 B1058.3   |
| F9 B5 B1051.6   |
| F9 B5 B1060.3   |
| F9 B5 B1049.7   |

All boosters that carried maximum payload have one thing in common:  
They all belong to F9 B5 group

# 2015 Launch Records

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| Month | Landing_Outcome      | Booster_Version | Launch_Site |
|-------|----------------------|-----------------|-------------|
| 01    | Failure (drone ship) | F9 v1.1 B1012   | CCAFS LC-40 |
| 04    | Failure (drone ship) | F9 v1.1 B1015   | CCAFS LC-40 |

There were only two launch records in 2015:  
Both belong to CCAFS LC-40 launch site  
Both failed in drone ship  
The first one was in January, while the second in April

## Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

Here is the statistics for given period.

It can be seen that landing outcome 'no attempt' occurs the most – 10 times.

Success and Failure in drone ship have same results, while the rest is <3.

| Landing_Outcome        | outcome_count |
|------------------------|---------------|
| No attempt             | 10            |
| Success (drone ship)   | 5             |
| Failure (drone ship)   | 5             |
| Success (ground pad)   | 3             |
| Controlled (ocean)     | 3             |
| Uncontrolled (ocean)   | 2             |
| Failure (parachute)    | 2             |
| Precluded (drone ship) | 1             |

A satellite view of Earth from space, showing the curvature of the planet and city lights at night. The image is a composite of a solid blue background on the left and a satellite photograph of Earth on the right. The Earth's surface is dark blue, with numerous bright yellow and orange lights representing cities and urban areas. The horizon line of the Earth is visible, separating the dark surface from the blackness of space.

Section 3

# Launch Sites Proximities Analysis

# The launch sites' locations on global map

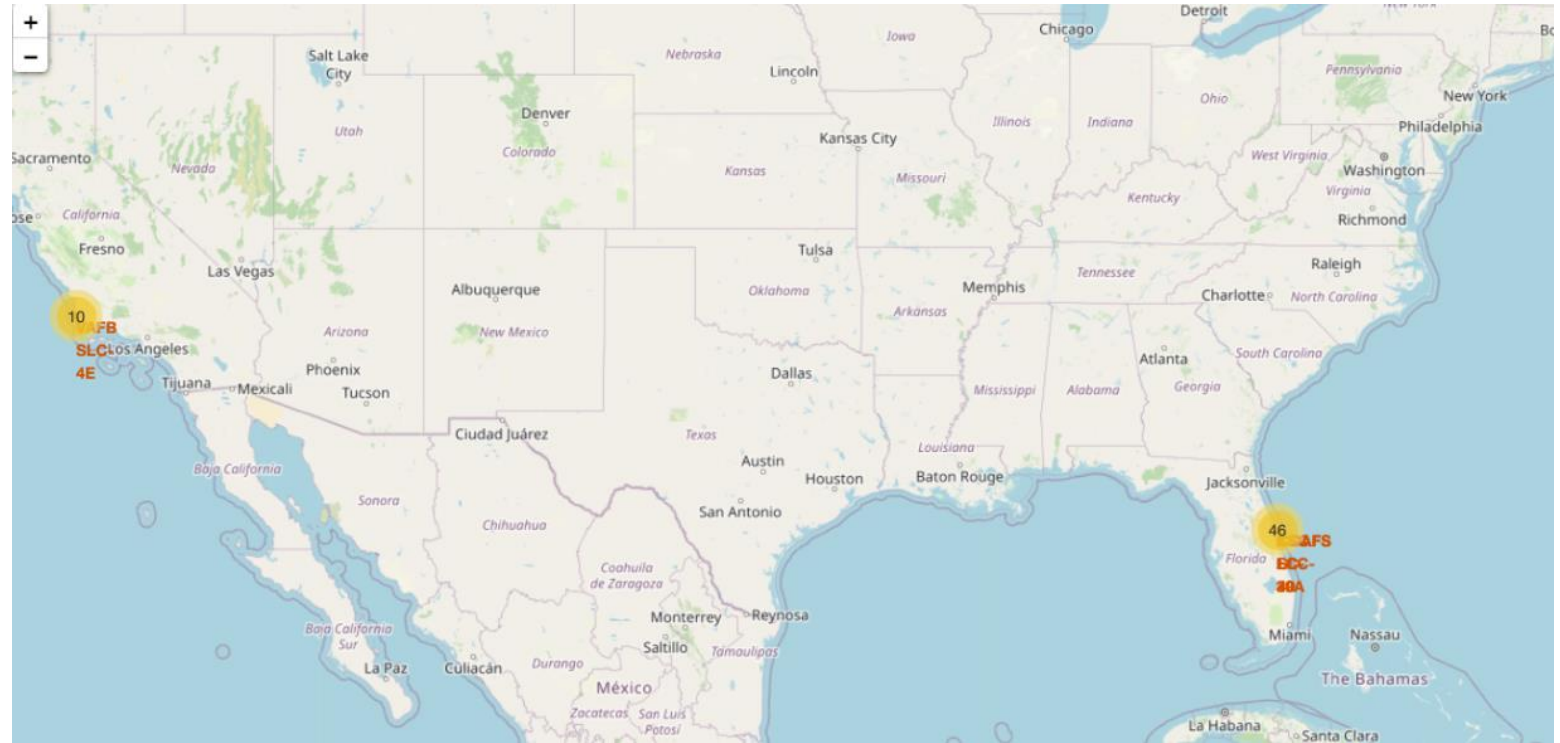
**Equator proximity:** U.S. launch sites are mid-latitude, not equatorial.

**Coastal proximity:** All sites are on or near coastlines to ensure safe launch trajectories.



# Number of launches on each site

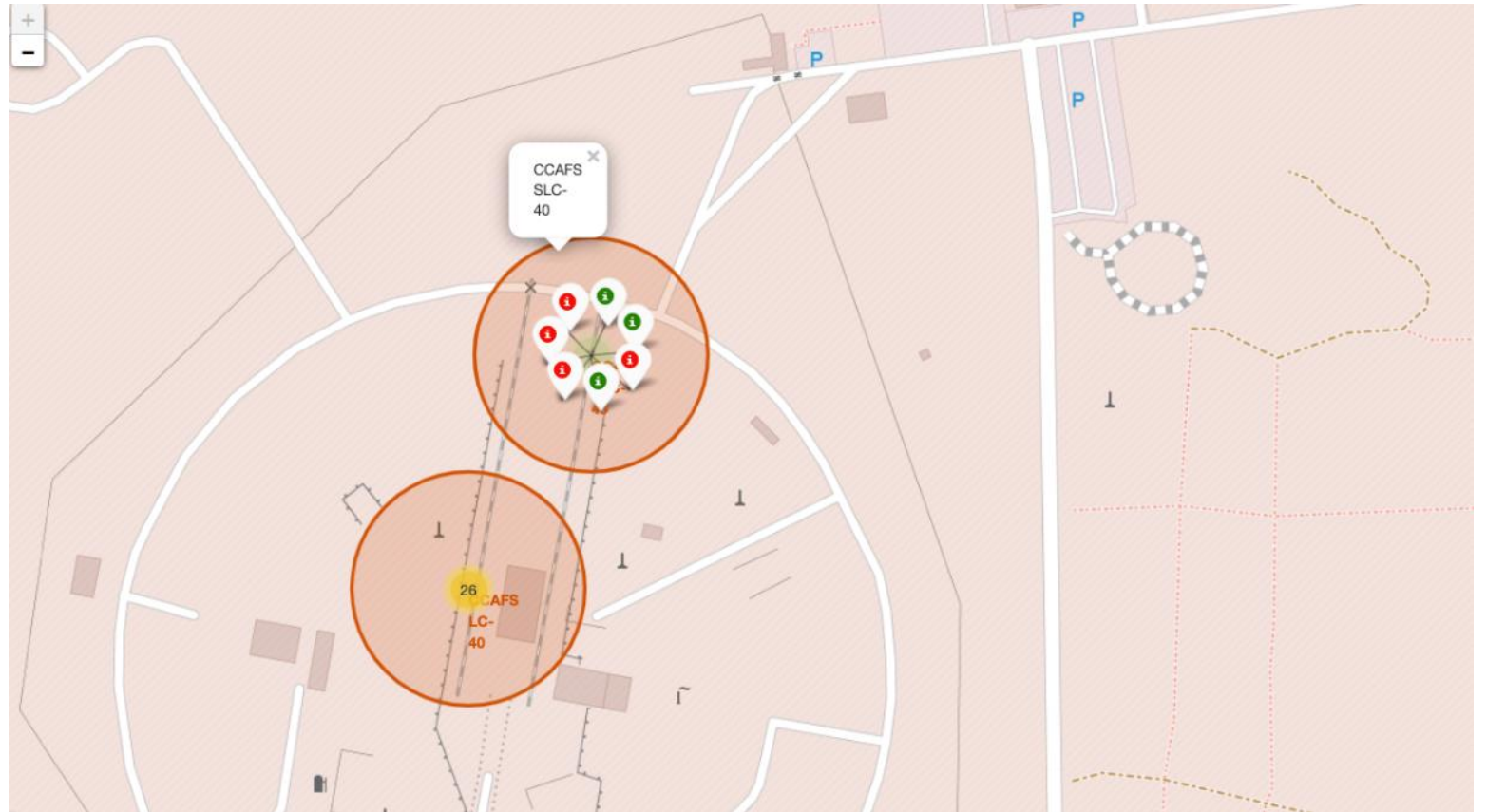
This map shows the number of flights on each launch site. There are 46 flights in Florida and 10 in California



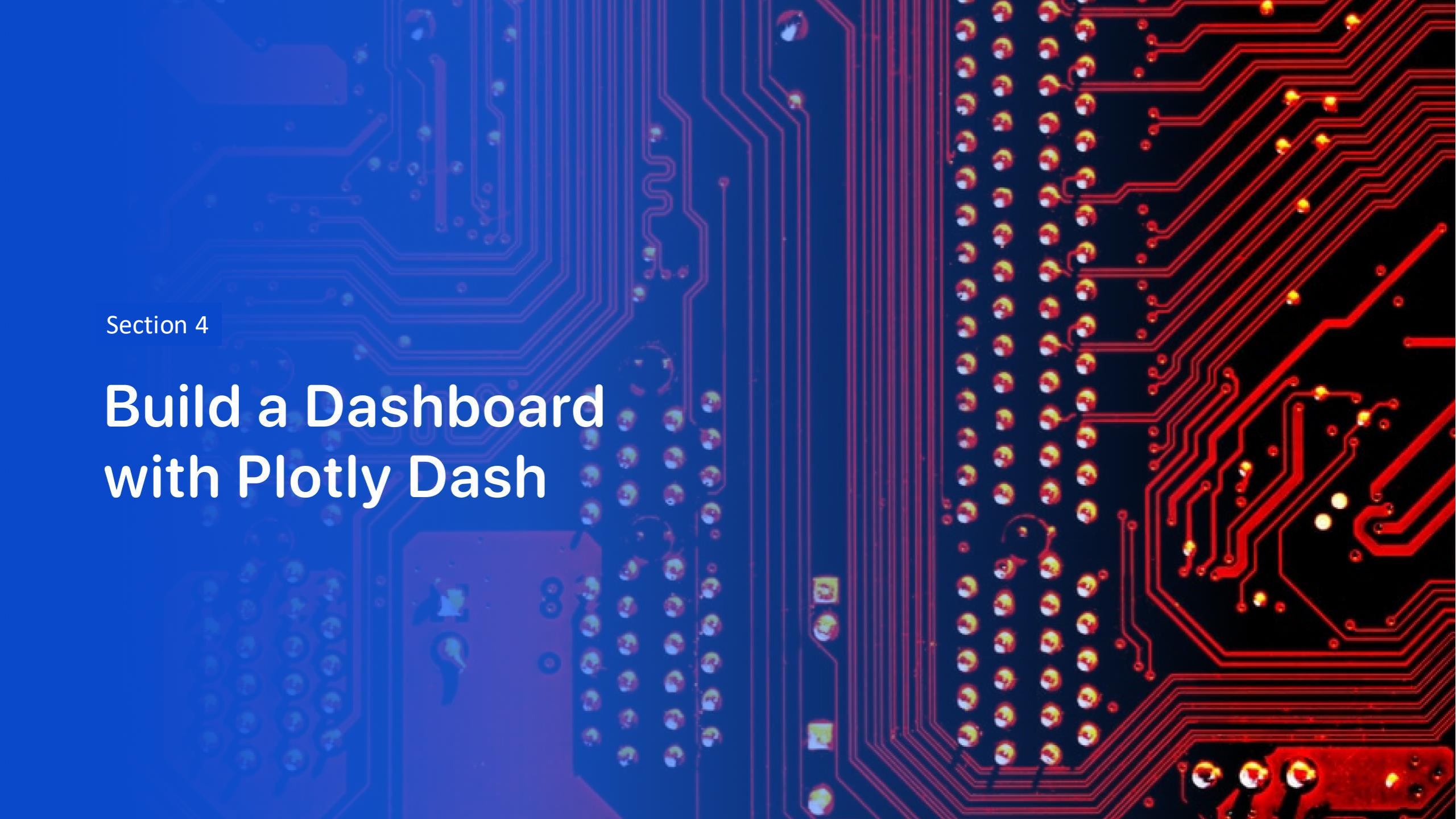


# The map of success rate of each launch site

The map indicates how many successful and unsuccessful flights have been on each site. As seen, there are 3 successful and 4 failure attempts in CCAFS SLC-40 launch site





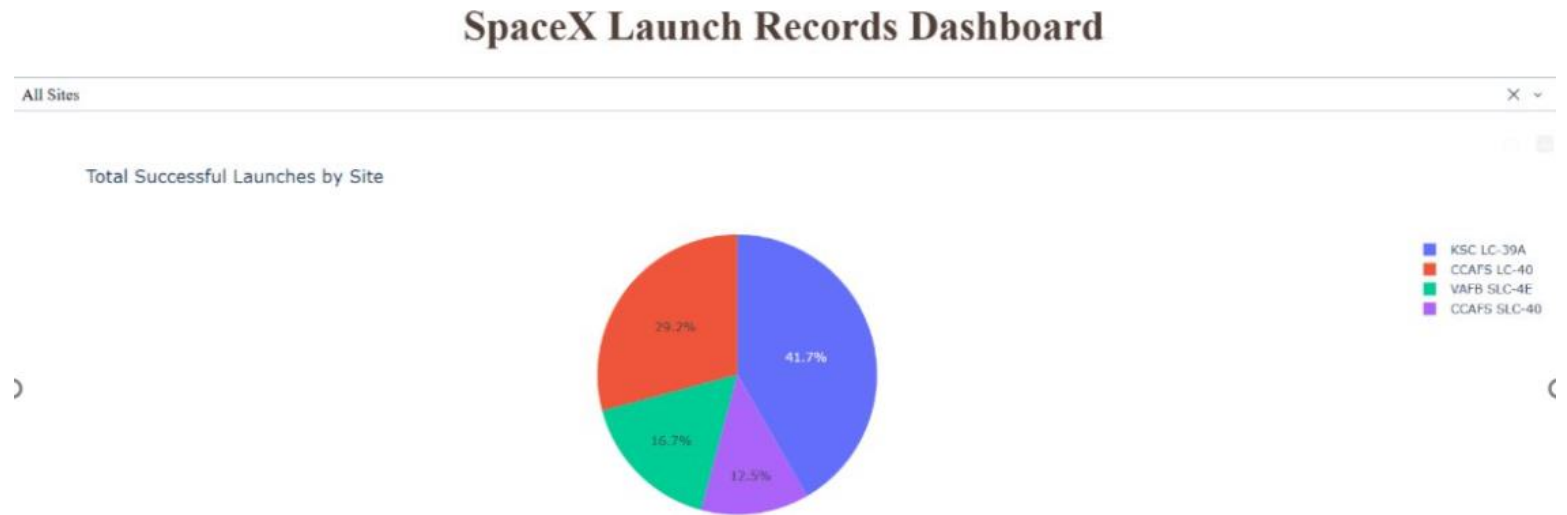


Section 4

# Build a Dashboard with Plotly Dash

# SpaceX launch records dashboard

This piechart indicates total successful launches by each site. The KLS LC-39A has highest success rate with 41.7% of total, while CCAFS SLC-40 has lowest with 12.5%



# Payload mass range VS Outcome

The scatter plot shows the relationship between Payload Mass range, which is about 4K to 8.5K. It indicates no relationship for v1.1 and FT booster versions

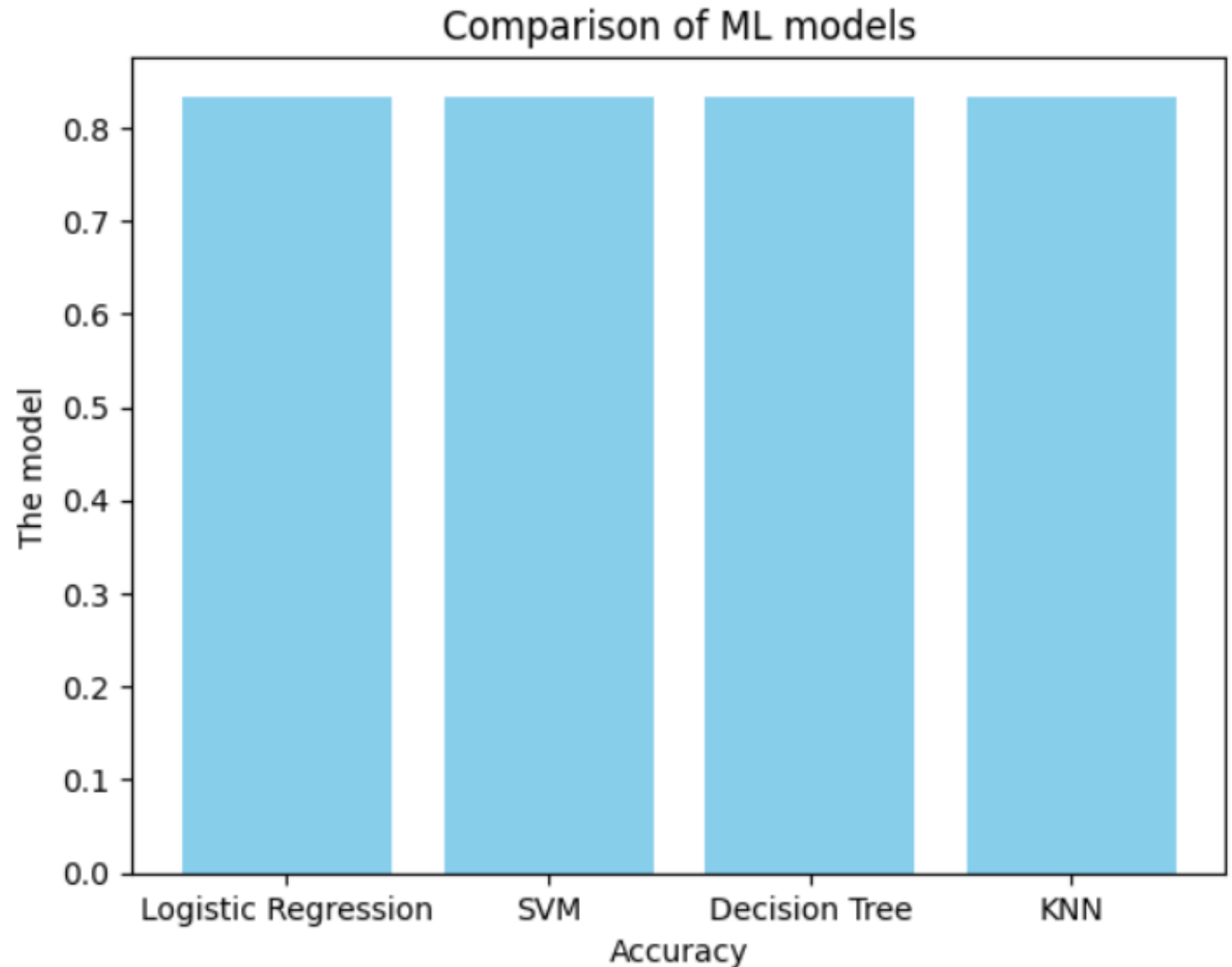


Section 5

# Predictive Analysis (Classification)

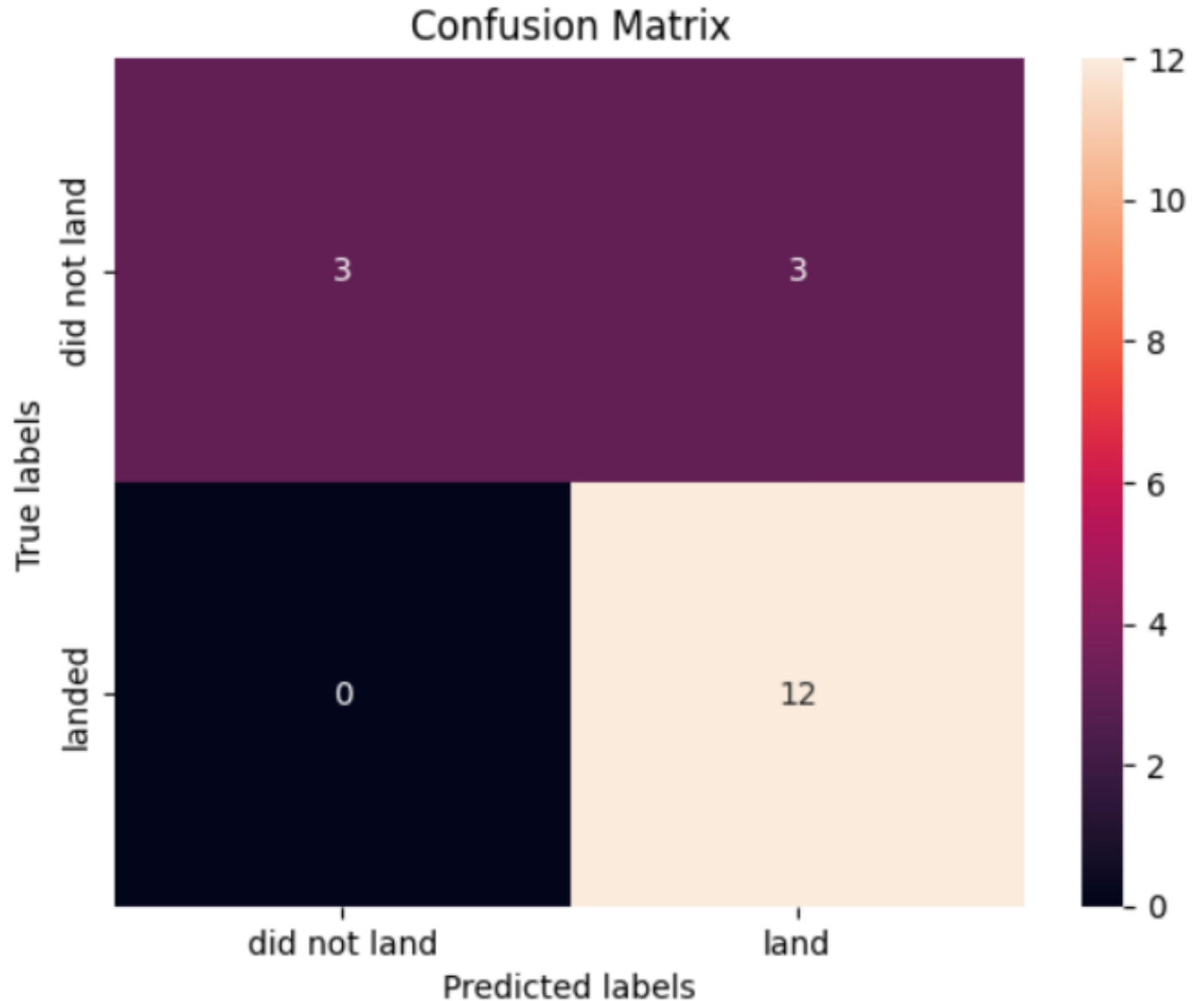
# Classification Accuracy

All classification models got the same accuracy.  
However, the decision tree classifier was a little bit overfitting



# Confusion Matrix

As it was mentioned before, all models performed the same, except Decision tree classifier. According to confusion matrix, all models have problems with false positive





# Conclusions

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- Problems That Got Answers
- Through this analysis, several key questions were addressed:
- **1. How did SpaceX's launch cadence evolve?**
- Early years (2010–2013): Few launches, mostly experimental.
- By 2020: Achieved **monthly cadence**, with multiple boosters reused up to **12+ times**.
- **2. What trends emerged in payload mass and orbit types?**
- Payloads grew from **~500 kg (ISS resupply)** to **>15,000 kg (Starlink mega-constellation)**.
- Orbit diversity expanded:
- **LEO/ISS** → dominant early missions.
- **GTO** → commercial satellites.
- **Polar/SSO** → specialized missions (Iridium, SAOCOM).
- **VLEO** → Starlink became the largest payload driver post-2019.
- **3. Did booster reusability improve reliability and efficiency?**
- Early boosters: 0–1 reuse.
- Block 5 boosters: up to **13 flights** (B1049, B1051, B1058).
- Landing success rate improved from **<50% (2010–2014)** to **>90% post-2017**.
- **4. Which launch sites and landing pads were most utilized?**
- **CCAFS SLC-40 (Cape Canaveral)**: Primary site for ISS and commercial launches.
- **KSC LC-39A (Kennedy)**: Crewed missions and heavy payloads.
- **VAFB SLC-4E (Vandenberg)**: Polar/SSO launches.
- Landing pads: **ASDS (drone ships)** became critical for GTO and Starlink missions, while **RTLS (ground pad)** supported ISS and LEO missions
- **5. What strategic milestones defined SpaceX's trajectory?**
- 2015: First drone ship landing attempts.
- 2016: First successful ASDS recovery.
- 2017: First booster reuse (SES-10).
- 2018: Block 5 introduction (optimized for reusability).
- 2019–2020: Starlink program drove record payloads and booster reuses.
- 2020: Crewed missions (Crew Dragon) marked entry into human spaceflight.



# Appendix

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- Sources:
- [https://en.wikipedia.org/wiki/List\\_of\\_Falcon\\_9\\_and\\_Falcon\\_Heavy\\_launches](https://en.wikipedia.org/wiki/List_of_Falcon_9_and_Falcon_Heavy_launches)

Thank you!

